

Human Microtubule-Associated Protein 1a (MAP1A) Gene: Genomic Organization, cDNA Sequence, and Developmental- and Tissue-Specific Expression

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Microtubule-associated proteins (MAPs) regulate microtubule stability and play critical roles in neuronal development and the balance between neuronal plasticity and rigidity. MAP1a (HGMW-approved symbol MAP1A) stabilizes microtubules in postnatal axons. We describe human MAP1a's genomic organization and deduced cDNA and amino acid sequences. MAP1a is a single-copy gene spanning 10.5 kb. MAP1a coding sequence is contained in five exons. Translation begins in exon 3. Human MAP1a contains 2805 amino acids (predicted molecular weight 306.5 kDa) and is slightly larger than rat MAP1a (2774 amino acids). Like rat and bovine MAP1a, human MAP1a contains conserved tubulin binding motifs in the amino-terminal region. The carboxy-terminal portion contains a conserved pentadecapeptide that is present in the light chain portion of rat and bovine MAP1a/LC2 polyprotein. We show that human MAP1a gene expression occurs almost exclusively in the brain and that there is approximately 10-fold greater gene expression in adult brain compared to fetal brain. Strong, interspecies conservation between human and rat MAP1a cDNA and amino acid sequences indicates important relationships between MAP1a's function and its primary amino acid sequence. © 1996 Academic Press, Inc.

INTRODUCTION

Development and maintenance of neuronal shape and the balance between neuronal rigidity and plasticity depend to a large extent on regulated assembly and disassembly of microtubules. Nascent microtubules are relatively unstable, having a tendency to depolymerize into α and β tubulin monomers (Baas *et al.*, 1993). Microtubule stability is controlled in part by microtu-

bule-associated proteins (MAPs), functionally related structural proteins that bind directly to microtubules (see Shiomura and Hirokawa, 1987, for example and review by Matus, 1988). In addition to microtubules, MAPs bind to actin and neurofilaments either alone or as cross bridges between microtubules and these cytoskeletal elements (see Pedrotti *et al.*, 1994, for example, and review by Matus, 1988). Phosphorylation of MAPs by specific type II cAMP-dependent and Ca^{2+} /calmodulin-dependent protein kinases affects the relative binding of MAPs to microtubules (as opposed to actin and neurofilaments) and determines the degree to which MAPs facilitate microtubule assembly (see Murthy and Flavin, 1983; Nishida *et al.*, 1981, and review by Matus, 1988). Phosphorylation decreases MAP's affinity for microtubules and increases MAP's affinity for neurofilaments and actin.

MAPs are classified by size into those of high molecular weight (~ 350 kDa, MAP1a, MAP1b, and MAP2) and those of intermediate molecular weight (~ 60 kDa, MAP2c, tau) (Matus, 1988). There are additional MAPs with molecular weights between 60 and 350 kDa (MAP3 and MAP4) and less well characterized MAPs smaller than 60 kDa (Matus, 1988). MAP1a and MAP1b are composed of a single heavy chain (in rat, MAP1a: 275 kDa, MAP1b: 244 kDa) and more than one light chain (LC) (24 to 27 kDa) (Vallee and Davis, 1983). LC1 and LC2 are encoded in the 3' ends of MAP1b and MAP1a mRNA (respectively) and produced by posttranslational cleavage of MAP1a/LC2 and MAP1b/LC1 polyproteins (Hammarback *et al.*, 1991; Langkopf *et al.*, 1992a). LC3 is the product of a separate gene (Mann and Hammarback, 1994). LC1 and LC3 bind to both MAP1a and MAP1b heavy chains (Kuznetsov and Gelfand, 1987). LC2 binds only to MAP1a heavy chain (Mann and Hammerback, 1994). Whereas microtubule binding domains of MAP1a and MAP1b are created by multiple subunits (amino-terminal portions of MAP1a and MAP1b heavy chains interacting with LC1, LC2, and LC3), the microtubule binding domains of MAP2, MAP4, and tau consist of discrete regions of a single protein (Hammarback *et al.*, 1991; Kuznetsov *et al.*, 1986; Schoenfeld *et al.*, 1989).

Sequence data from this article have been deposited with the EMBL/GenBank Data Libraries under Accession Nos. U38291 and U38292.

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Every MAP that has been examined has shown significant changes in abundance and occasionally in structure at different developmental stages (see Francon *et al.*, 1982, for example and review by Matus, 1988). In the mouse, MAP2b, MAP5, MAP2c, MAP3a, MAP3b, and juvenile tau are abundant in the embryo and newborn periods and then gradually decrease to adult levels by Day 20 (except MAP2b, which does not decrease postnatally) (Matus, 1988). In contrast, the concentrations of MAP1a, MAP2a, and adult tau are very low in the embryonic period and increase postnatally. In murine brain, for example, MAP1a protein is present at very low levels during the embryonic period. Beginning 48 h before birth, MAP1a protein increases, reaching a maximum level at approximately Day 20, and remains relatively constant thereafter (Matus, 1988). Developmental changes in MAP abundance correlate with stages of neurite growth. MAP1a (detectable shortly before birth) and MAP2a (not detectable prenatally) accumulation increases in the postnatal period as neuronal processes are growing and reaches a plateau at the end of the major phase of axon and dendrite formation (Matus, 1988). In contrast, expression of MAP1b and MAP3 begin in the embryonic period and end between Postnatal Days 10 and 20, coinciding with the transition from the major period of axonal growth to axonal maintenance (Matus, 1988). These changes in patterns of expression suggest that MAP1b, MAP2, and MAP3 are involved in formation of neuronal cytoskeleton and that MAP1a and MAP2a are involved in maintenance of the neuronal cytoskeleton (Mann and Hammerback, 1994).

MAP1a is the largest MAP and is localized predominantly in neuronal cell body, axons, and dendrites (Matus, 1988). Lien *et al.* (1994) reported that human expressed sequences (EST) 00368, EST 01683, and EST 01974 had 80, 87, and 95% (respectively) amino acid homology with rat MAP1a. We used these ESTs to identify a bacteriophage P1 containing the entire human MAP1a² genomic sequence. In this report we describe human MAP1a's genomic organization and cDNA and amino acid sequences and show that MAP1a is a single-copy gene whose expression is confined largely to the postnatal central nervous system.

MATERIALS AND METHODS

Isolating, sequencing, and amplifying MAP1a genomic DNA. Four plasmids [HHCJ80 (1.7-kb insert), HIBBBI94 (2.6-kb insert), HHC PF54 (1.4-kb insert), and HIBBB73 (1.9-kb insert)] containing human ESTs 00368, 01683, 01974, and 06170 (respectively) were obtained from American Type Tissue Collection (ATCC; Rockville) and sequenced completely in both directions using automated sequencing. DNA sequencing was performed using the PRISM Ready Reaction DyeDeoxy terminator cycle sequencing kits (Perkin-Elmer), following the protocol (Part No. 401388) and using Centri-Sep spin columns (Princeton Separations). Samples were electrophoresed on 6% polyacrylamide gels and DNA sequences were determined by an Applied Biosystems 373A DNA sequencer. Each plasmid was

sequenced by a series of sequencing primers. Overlapping sequences were assembled into a contiguous DNA sequence by using the Sequencher program (GeneCodes, Ann Arbor, MI). Overall, automated DNA sequencing resulted in definite nucleotide identification for more than 90% of the DNA sequences. Less than 10% of nucleotide positions were ambiguous (identified as unknown, purine, or pyrimidine, for example). Most of these latter ambiguities were resolved by improved quality of sequence in overlapping fragments and by sequencing the opposite strand.

PCR primers to EST 01974 (Table 1, primers 1A and 1B), EST 00368 (Table 1, primers 2A and 2B), and EST 01683 (Table 1, primers 3A and 3B) were used to screen a human genomic DNA (gDNA) library cloned in bacteriophage P1. An 85-kb P1 clone (Dupont Merck Pharmaceutical Co., HHF#1-374F3) containing ESTs 00368, 01683, and 01974 was identified, digested with *EcoRI*, and subcloned (L. Meyer, Genome Systems Inc., St. Louis, MO) into KS- Bluescript (Stratagene, La Jolla, CA). A 10.2-kb subclone clone (designated G5) was shown by PCR to contain ESTs 00368, 01683, and 01974 and was sequenced completely in both directions. Aligning sequences of plasmids containing partial MAP1a cDNA (ATTC clones HHCJ80, HIBBBI94, HHC PF54, and HIBBB73) with gDNA P1 subclone G5 (Fig. 1) indicated that G5 contained the entire genomic MAP1a sequence except 80 bp of coding sequence in the 5' portion of HHCJ80 and 928 bp of coding sequence in the 3' end of plasmid HIBBB73. Oligonucleotide primers to amplify the 5' end of HHCJ80 (Table 1, primers 4A and 4B) and the 3' end of plasmid HIBBB73 (Table 1, primers 5A and 5B) were designed and used to identify P1 subclones B7 and 34 containing (respectively) the 5' and 3' ends of the MAP1a coding sequence.

Genomic organization. Intron/exon boundaries were determined by comparing gDNA and cDNA sequences for regions in which available cDNA sequences (plasmids HHCJ80, HHC PF4, HFBIC146, and HIBBB73) overlapped gDNA sequences (plasmids B7, G5, and 34) (Fig. 1). Four introns totaling 795 bp were found in this way. We used two methods to determine if there were small introns in the gaps between MAP1a cDNA clones HHCJ80 and HIBBBI94 and between HIBBBI94 and HHC PF54 (Fig. 1), areas in which MAP1a gDNA was not spanned by available cDNA fragments. First, these gaps were amplified (Table 1, primers 6 through 14) as overlapping fragments (Fig. 2A) from adult brain-derived cDNA (Catalog No. 936211, Stratagene) and compared to amplifications using MAP1a gDNA template (P1 HHF#1-374F3; Dupont Merck). Similar sizes of amplification products from cDNA and gDNA templates (Fig. 2A) indicated that there were no introns between cDNA fragments in HHCJ80 and HIBBBI94 and no introns 2444 bp distal to the 3' end of HIBBBI94. A 730-bp region (between G5 positions 7854 and 8584) could not be analyzed readily by this approach. Therefore, ribonuclease protection assay was used to determine whether there were small introns in this gap. This region was amplified (Table 1, primers 15 through 18) (Fig. 2B) as two fragments from MAP1a gDNA in P1 HHF#1-374F3. Inclusion of the T7 promoter sequence (5' to 3': CCAAGCTTCTAA-TACGACT) in the 3' end of reverse PCR primers allowed amplification products to serve as templates for transcription of RNA probes (Logel *et al.*, 1992). Following gel purification of amplified probe templates, RNA probes were prepared, hybridized with adult brain mRNA (Clontech, Palo Alto, CA), digested with RNases, and electrophoresed according to the manufacturer's protocol (RPA II, Ambion, Austin, TX).

Southern Blot analysis. Probe was prepared by random prime labeling the 700-bp *EcoRI* insert of P1 subclone B7 (Feinberg and Vogelstein, 1984). B7 was chosen because it contains the 5' end of MAP1a coding sequence. Human genomic DNA (10 μ g) was digested with *Bam*HI and *Kpn*I, electrophoresed on 1% agarose gel, transferred to nylon membrane, probed, and washed according to standard procedures.

Northern Blot analysis. Multiple tissue Northern blots (Stratagene) and human fetal brain Northern blots (Stratagene) were analyzed according to the manufacturer's guidelines. Probe for Northern blot analysis was prepared by amplifying a 250-bp portion of MAP1a cDNA plasmid HHCJ80 followed by gel purification and random prime labeling (Feinberg and Vogelstein, 1984).

² The HGMW-approved symbol for the gene described in this paper is MAP1A.

TABLE 1

Oligonucleotide Primers and PCR Conditions for Identifying and Analyzing Human MAP1a

Purpose	Primer	Oligonucleotide (5' → 3')	PCR conditions ^a	Cycling parameters ^b
P1 isolation	1A	AGCTCCCTGCTGCTTGACACAGTC	A	1
	1B	GGTGGTGATTTGAAGCCCTTGCTCTT		
	2A	CTGTGCCTTACCAACCTCAAGCCCA		1
	2B	TCCCGTTTGGCCAACTTGCTCAC		
	3A	CCACTGACCACATAGTAGGATGCA		1
	3B	GGACCCAATGGCCTTGAGTTCCAAGG		
Identifying 5' end of MAP1a gDNA	4A	TGCCACCAGAGTGAGATCCTAG		2
	4B	TAATACGACTCACTATAGGG		
Identifying 3' end of MAP1a gDNA	5A	AAGGGAGAAGCTGAGCCACA		2
	5B	TTGGAGGTGGCTCCGTTAC		
Comparing MAP1 gDNA and cDNA Fig. 2a	6A	GGAGAGCCTCMAAAGCCACTAC	B	3
	6B	CCCTCTCCTCACGCTTCA		
	Fragment 2	7A	B	3
		7B		
	Fragment 3	8A	B	4
		8B		
	Fragment 4	9A	B	3
		9B		
	Fragment 5	10A	B	3
		10B		
	Fragment 6	11A	B	3
		11B		
	Fragment 7	12A	B	1
		12B		
	Fragment 8	13A	B	6
		13B		
	Fragment 9	14A	B	5
		14B		
	Fragment 10	15A	B	3
		15B		
	Fragment 11	16A	B	3
		16B		
Ribonuclease protection assay (See text for 19-bp T7 promoter sequence added to 5' end)	17A	CAATGGCCCAACTGAAACC		7
	17B	GGGGGTGTCTCATCAGTC		
	18A	GTTCGGCTAGTAGGAAGAG		7
	18B	TGAGCGTTTTCCCGAAG		

^a PCR conditions were 10 mM Tris, pH 8.3, 50 mM KCl, 0.8 μ M each primer, 200 μ M each dNTP, 1.5 mM MgCl₂, 1U *Taq* polymerase except as noted: condition A, 1.75 mM MgCl₂; condition B, 1% Tween 20 (used for cDNA, PCR, only).

^b PCR cycling parameters were: (1) 94°C (30 s), 55°C (30 s), 72°C (1 min), 35 cycles; (2) 94°C (30 s), 58°C (30 s), 72°C (30 s), 30 cycles; (3) 94°C (30 s), 58°C (30 s), 72°C (30 s), 35 cycles; (4) 94°C (45 s), 64°C (1 min), 72°C (45 s), 5 cycles, 94°C (45 s), 60°C (45 s), 72°C (45 s), 30 cycles; (5) 94°C (45 s), 55°C (45 s), 72°C (45 s), 30 cycles; (6) 94°C (30 s), 57°C (30 s), 72°C (30 s), 35 cycles; (7) 94°C (45 s), 60°C (45 s), 72°C (45 s), 30 cycles.

RESULTS AND DISCUSSION

MAP1a is a single-copy gene. MAPs are functionally and structurally related peptides. Conservation of amino acid sequences between MAPs of different species (Lien *et al.*, 1994) and of different classes suggests that individual MAP genes (such as MAP1a and MAP1b) may have evolved by duplication of an archival

MAP gene. We performed Southern blotting to determine whether MAP1a was a single-copy gene or whether there were multiple genes or pseudogenes. Southern blot (Fig. 3) of *Bam*HI- and *Kpn*I-digested gDNA each resulted in a single band (10.5 and 5.6 kb, respectively). Lack of variation of these restriction enzyme sites in MAP1a's 5' flanking region indicates that MAP1a is a single-copy gene.

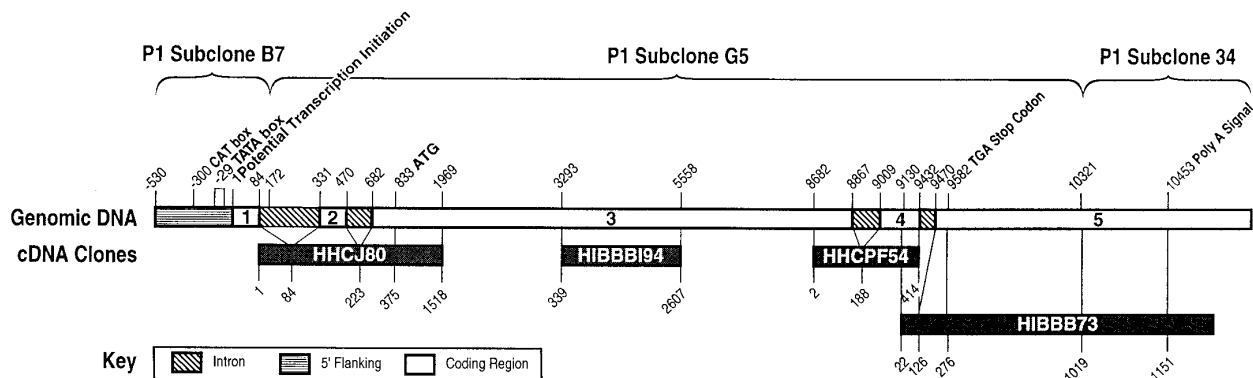


FIG. 1. Genomic organization of human microtubule-associated protein 1A (MAP1a). An 85 kb P1 clone (Dupont Merck Pharmaceutical Co., HHF#1-374F3) containing human EST 00368, EST 01683, and EST 01974 was identified and subcloned, and plasmid subclones (B7, G5, and 34) containing these ESTs were isolated and sequenced completely. Human cDNA plasmids (HHCJ80, HIBBBI94, HHCPI54, and HFBBB73) containing these ESTs were obtained from ATTC and sequenced completely. Intron/exon boundaries were determined as described in the text (see Figs. 2A and 2B).

MAP1a genomic organization and cDNA and amino acid sequences. We deduced MAP1a cDNA sequence by sequencing MAP1a genomic DNA (contiguous P1 subclones B7, G5, and 34) and eliminating introns [genomic DNA regions (totaling 795 bp) not found in cDNA sequence (plasmids HHCJ80, HIBBBI94, HHCPI54, and HFBBB73)]. To determine whether there were introns in gDNA regions not overlapped by available cDNA fragments, we compared the sizes of PCR products from gDNA template with those of cDNA templates and used the ribonuclease protection assay to examine a small region that was not readily amplified. PCR product sizes obtained from gDNA were extremely similar to those from cDNA templates (Fig. 2A). Ribonuclease protection assays showed that human brain-derived RNA contained species that fully protected the riboprobe (created from MAP1a gDNA template). The 19-bp differences in sizes of riboprobes alone [Fig. 2B, fragments A (442 bp) and B (465 bp)] and the sizes of the RNA-protected riboprobes [Fig. 2B, fragments A (423 bp) and B (446 bp)] are due to the 19-bp T7 promoter sequence incorporated into riboprobes that is not present in MAP1a mRNA. Ribonuclease protection assay and comparison of PCR product sizes from gDNA and cDNA templates showed no evidence of introns in gDNA regions not overlapped by available cDNA fragments.

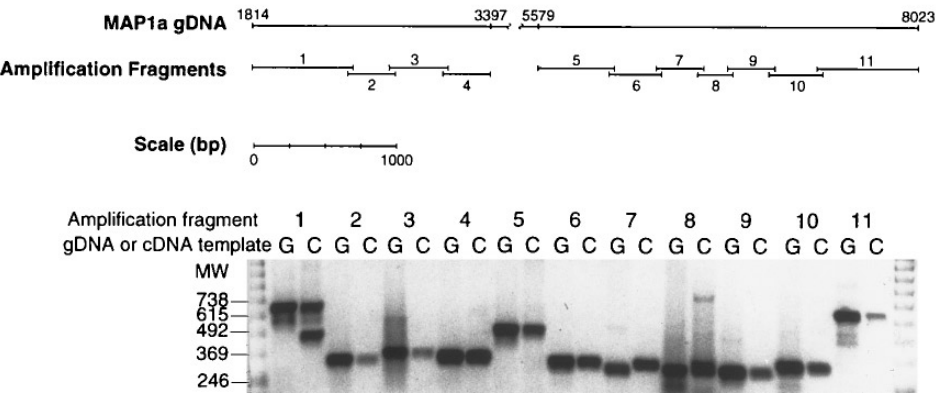
The MAP1a gene (TATA box to polyadenylation signal) spans 10,482 kb and is divided into four introns and five exons. The probable translation initiation (first ATG) of human MAP1a showed complete amino acid

agreement with rat MAP1a's translation initiation (MDGVAEFS. . .; Langkopf *et al.*, 1992). Translation (first ATG) begins in the third exon and terminates (first in-frame TGA) in the fifth exon. Table 2 shows the location of consensus promoter elements (TATA and CAAT), translation initiation, translation termination polyadenylation signal, and location of each exon and intron. We designated the 5' end of MAP1a cDNA plasmid HHCJ80 as position one. Although only 28 bp downstream from the TATA box, it is possible that transcription actually begins slightly upstream from our designated position one. MAP1a's complete genomic sequence (Accession No. U38291) and cDNA sequence (Accession No. U38292) have been deposited with GenBank.

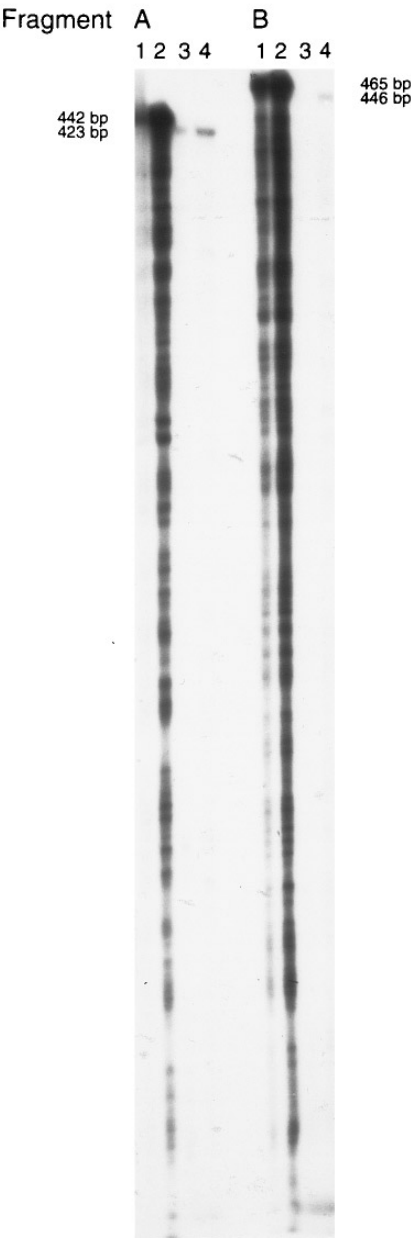
The coding sequence (first ATG to stop codon) measures 8418 bp and predicts a polypeptide (2805 residues) of 306.5 kDa. As noted above, although automated DNA sequencing provided very-high quality sequence, 73 bp (0.9%) of MAP1a cDNA positions remain ambiguous. As a consequence, the identity of 72 amino acids (2.6%) could not be predicted (Fig. 4, indicated with an "x"). These ambiguous amino acid residues were scattered throughout MAP1a's sequence. There is extremely strong conservation between human and rat MAP1a predicted amino acid sequences (Langkopf *et al.*, 1992b). This is particularly true of the amino-terminal region. The first 500 amino acids show 94% amino acid identity between human and rat MAP1a (Langkopf *et al.*, 1992b). This region contains putative microtubule binding domains described below. The region

FIG. 2. Determining the absence of introns in MAP1a gDNA regions not spanned by cDNA. (A) MAP1a regions spanning positions 1814 through 3397 (fragments 1 through 4) and 5582 through 8026 (fragments 5 through 11) were amplified (see Table 1) from gDNA (lanes "G") and cDNA (lanes "C") templates, and product sizes were compared by agarose gel electrophoresis. (B) Ribonuclease protection assays. Overlapping RNA probes to MAP1a gDNA positions 7909 through 8331 (fragment A) and 8243 through 8719 (fragment B) were created by amplifying portions of MAP1a gDNA template (P1 subclone G5; see Table 1) using reverse primers that included T7 promoter sequences in their 5' ends and transcribing and ³²P labeling the RNA products from these templates. These riboprobes were used to protect adult human brain-derived RNA samples from RNase digestion and were then electrophoresed on a 5% polyacrylamide/8 M urea. Lanes 1 through 4 contain (respectively) RNA probe alone, RNA probe and yeast tRNA without RNase treatment, RNA probe and yeast tRNA followed by RNase treatment, and RNA probe and adult brain RNA followed by RNase treatment.

A



B



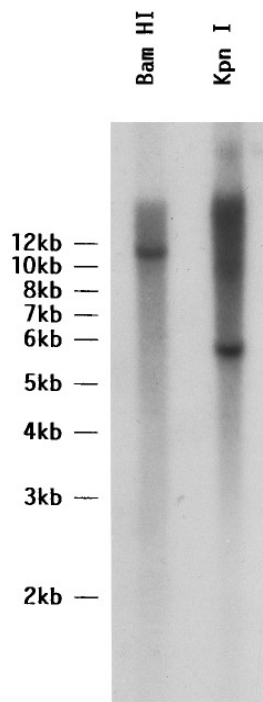


FIG. 3. Southern blot analysis shows that human MAP1a is a single-copy gene. Human genomic DNA was digested with *Bam*HI (lane 1) and *Kpn*I (lane 2), electrophoresed on 1% agarose gel, transferred to nylon membrane, probed, and washed according to standard procedures (Sambrook *et al.*, 1989). Probe was prepared by random prime labeling the insert of P1 subclone B7, which contains the 5' end of MAP1a gDNA.

of interspecies amino acid conservation extends well beyond this region, however, with the first 1370 amino acids showing 88% identity between human and rat. Following this region, human MAP1a contains a region of approximately 300 amino acids that shows much less interspecies conservation. (We carefully examined this region in multiple reading frames but could not substantially improve alignment to rat MAP1a.) The remaining region of approximately 1100 amino acids again shows extremely strong conservation (80%) between human and rat MAP1a.

Microtubule binding in MAP1b is thought to be mediated in part by amino terminal clusters of [K/R] [K/R] [D/E] motifs (Noble *et al.*, 1989). Such motifs are also present in rat MAP1a (Langkopf *et al.*, 1992b). The presence of eight [K/R] [K/R] [D/E] motifs in human MAP1a's amino-terminal region (Fig. 4) suggests that this region (including amino acids 415 through 541) also participates in microtubule binding. The functional significance of two isolated [K/R] [K/R] [D/E] motifs found in MAP1a's central portion (Fig. 4) is not clear. As noted previously, the rat MAP1a gene encodes both a heavy chain and a light chain (LC2). A pentadecapeptide (PEGLSSSESGRVERLR) found in bovine and rat LC2 is also present in human MAP1a ("PEGLSSSESGRxERLR") beginning at residue 2582 (Fig. 4) (Langkopf *et al.*, 1992b). This suggests that LC2 is encoded in the carboxy-terminal portion of human MAP1a cDNA as it is in other species.

Recently, Fukuyama and Rapoport determined the sequence of 1575 bp of human MAP1a cDNA (Fukuyama and Rapoport, 1995). There is excellent (96%) agreement between our MAP1a cDNA sequence (positions 4447 through 5977) and the first 1509 bp of Fukuyama and Rapoport's MAP1a sequence. The remaining 39 bp of their MAP1a cDNA sequence does not agree well with sequences we obtained from MAP1a gDNA (in P1 subclone G5).

MAP1a and MAP1b are structurally related peptides. As noted above, microtubule binding domains in the amino-terminal region are shared between human MAP1a and MAP1b. Pustell matrix analysis (data not shown) indicates that human MAP1a and human MAP1b have moderate conservation of amino acid sequences in both amino- and carboxy-terminal regions (data not shown). Our Pustell matrix comparison of human MAP1a and MAP1b is thus very similar to that reported by Langkopf *et al.* (1992b) for rat MAP1a and MAP1b. Any functional differences in MAP1a's and MAP1b's effects on microtubule stability are thus probably related to divergent sequences in the central portion of each peptide. In addition to the different number of exons (five in MAP1a, seven in MAP1b), the striking

TABLE 2

Human MAP1a Genomic Organization

CAAT	gDNA position	Sequences flanking exons (5' to 3')
CAAT	-300	
TATA	-29	
Translation initiation (first ATG)	833	
Stop codon (TGA)	9,582	
Poly(A) signal (AATAA)	10,453	
Exon 1	1-84	5' flank: TCCTACATCTCTCCCTCAGG 3' flank: AAGGCCAGGGCCTGTGCCTT
Intron 1	85-331	
Exon 2	332-470	5' flank: CCCCNTCTNCCATTCCAGGT 3' flank: GTAAGTTCCATGCCAGAGTG
Intron 2	471-682	
Exon 3	683-8867	5' flank: TCTATCTCCTATTCTTCTAG 3' flank: GTAAGTATATCATGNAAWCT
Intron 3	8868-9009	
Exon 4	9010-9234	5' flank: ATCCTGGCTTGTCTCTACAG 3' flank: GTAGCAAAGGACACCAAGGA
Intron 4	9235-9432	
Exon 5	9433-	5' flank: CTTATCTCCCTTCTAGGTGA Sequences flanking (3') poly(A) signal: GCCACTGNCTGACATCCCAA

[illegible]

FIG. 4. Human MAP1a deduced amino acid sequence.

difference between human MAP1a's and MAP1b's genomic organization is the size of introns. Whereas MAP1a spans 10.5 kb, MAP1b spans 101 kb of genomic DNA (Langkopf *et al.*, 1992b). Like human and rat MAP1b (Lien *et al.*, 1994), the majority (73%) of the total human MAP1a gene is contained in a single large exon (exon 3: 8.2 kb). Despite similarity of the sizes of human MAP1a exons 1 (83 bp), 2 (142 bp), and 4 (224 bp) and human MAP1b exons 2 (82 bp), 4 (140 bp), and 6 (238 bp) (respectively), homologous DNA sequences in these two genes do not appear to have the same intron/exon organization. For example, MAP1a sequences in exon 1 (83 bp) are not homologous to those in similarly sized MAP1b exon 2 (82 bp) but are homologous to those in MAP1b exons 3 and 4. While conservation of amino acid sequences between MAP1a and MAP1b indicates that these peptides may have arisen from an archival MAP gene, significant differences in genomic organization and large regions in which amino acid sequences diverge indicates the important role that genomic organization and primary amino acid sequences play in the specific functions and regulation of these unique proteins.

Human MAP1a exhibits developmental- and tissue-specific expression. Northern blot analysis showed that MAP1a mRNA expression is confined almost entirely to the brain (Fig. 5). Interestingly, a small amount

of MAP1a also appears to be transcribed in the adult heart; this has been observed previously (Fukuyama and Rappaport, 1995). There was no evidence of alternative processing of MAP1a mRNA (Fig. 5). As in other species, human MAP1a is expressed predominantly in adult brain (Fig. 5). We observed approximately 10-fold greater abundance of MAP1a mRNA in adult brain than in fetal brain. Like Fukuyama and Rappaport (1995), we observed relatively uniform abundance of MAP1a mRNA in different adult brain regions (Fig. 5). The size difference between human MAP1a transcript (10.4 kb) and MAP1a cDNA (8.4 kb) is attributed to a long 3'-untranslated region, similar to that reported for rat MAP1a (Langkopf *et al.*, 1992b).

Commercially available MAP1a cDNA plasmids contain extraneous sequences. We identified several spurious sequences in MAP1a cDNA plasmids obtained from ATCC. We attribute these cDNA sequences, which bear no similarity to human MAP1a gDNA or rat MAP1a cDNA (Langkopf *et al.*, 1992b), to artifacts of cDNA construction. For example, following reverse transcription, cDNAs were *EcoRI*-linkered, digested with *EcoRI*, and ligated into Lambda Zap (Stratagene). Therefore, cDNAs should not have internal *EcoRI* sites in cDNAs. Nonetheless, the first 338 bp of plasmid HIB-BBI94, which bears no similarity to human MAP1a gDNA or rat MAP1a cDNA, is flanked by *EcoRI* sites,

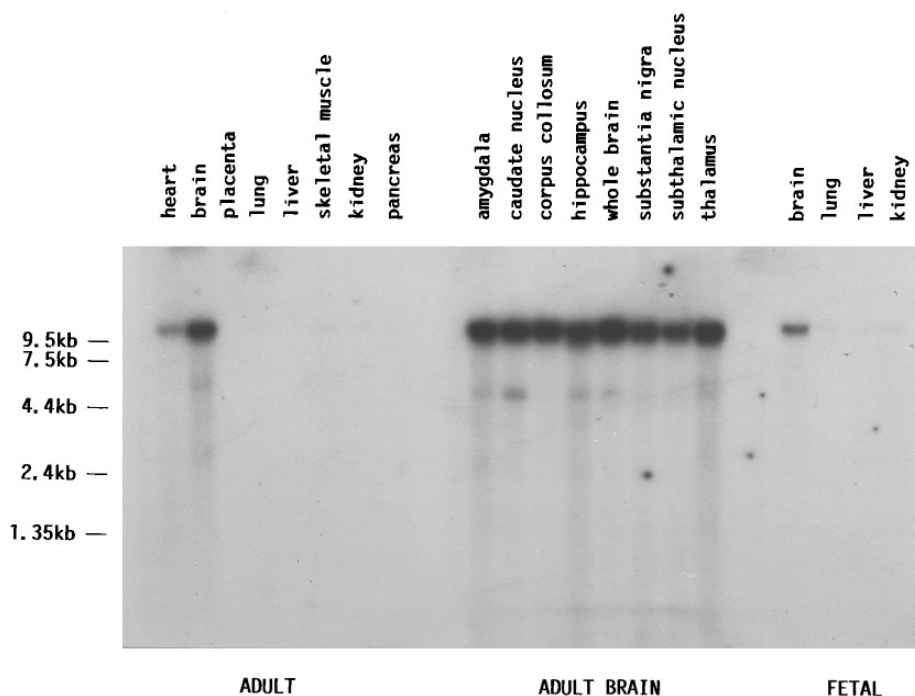


FIG. 5. Developmental- and tissue-specific transcription of human MAP1a. Northern blots prepared from multiple adult and fetal tissues and from different adult brain regions (Stratagene, LaJolla) were probed with a 250-bp portion of plasmid HHCJ80 (ATCC, Rockville).

indicating that two separate cDNAs were joined during the cloning process. Whereas the first 1518 bp of cDNA plasmid HHCJ80 agrees perfectly with gDNA, the terminal 132 bp fails to align with MAP1a gDNA. (We know that this region is not intronic because of comparison of PCR product sizes using cDNA and gDNA templates, described above.) It is interesting that the cDNA plasmid HHCPF54 aligns perfectly to gDNA until the beginning of intron 4. Extraneous sequences in the remainder of plasmid HHCPF54 (not present in MAP1a gDNA or in partially overlapping MAP1a cDNA plasmid HFB73) may have come from aberrant mRNA splicing and the inclusion of unrelated mRNA sequences.

Conclusions. MAPs play fundamental roles regulating development of neuronal shape during embryogenesis, maintenance of neuronal shape during adulthood, and changing neuronal shape during synaptic reorganization and following injury (Yin *et al.*, 1995; Blomgren *et al.*, 1995). Because of their central role in regulation of the neuronal cytoskeleton, it is reasonable to suspect that abnormal structure, function, or regulation of MAPs may contribute to developmental neurologic disorders such as Rett syndrome (Kaufmann *et al.*, 1995) and degenerative neurologic disorders including Alzheimer disease (Takahashi *et al.*, 1991). Abnormally phosphorylated Tau, for example, is present in neurofibrillary tangles in Alzheimer disease. In addition, MAP1b, whose expression is confined normally to fetal brain, is expressed in hippocampal neurons in brains from Alzheimer disease patients (Takahashi *et al.*, 1991). The relationship of this apparent indicator of neuronal sprouting to the progressive neuronal degen-

eration of Alzheimer's disease is not clear. Knowing the primary structure of individual MAPs provides the foundation for determining the basis for the diverse effects of individual MAPs on microtubule stability. Understanding the regulation of MAP gene transcription and identifying factors that control MAP post-translational processing (including phosphorylation and cleavage of MAP1a and MAP1b into heavy and light chains) will advance our understanding of the developmental regulation of the neuronal cytoskeleton and the relationship of these processes to developmental and degenerative neurologic disorders.

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